

Executive Summary

Uncertainty-Aware Survival Modeling of Pavement Deterioration with Connected Vehicle Data

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1. The Problem

Road agencies spend **millions of dollars** on pavement maintenance every year. They rely on models that predict *when* a road will need repair but these models typically give a **single number** (e.g., “this road will need work in 18 weeks”) without saying **how reliable that number is**.

If the prediction is off by just a few weeks, the agency either:

- **Intervenes too early** → wastes budget on roads that do not need repair yet
- **Intervenes too late** → faces higher rehabilitation costs and safety risks

The gap: Decision-makers need predictions with **confidence bounds**, not just point estimates. This thesis closes that gap.

2. Approach

The study analyzes weekly connected vehicle (CV) data from **3,707 Swedish road sections** across two municipalities (Nacka and Östersund) over 144 weeks (≈ 2.8 years), covering **3,357 deterioration events**. A deterioration event is defined as the first time a section’s International Roughness Index (IRI) crosses the intervention threshold of 2.5 m/km.

Three modeling components are used:

1. **Kaplan–Meier curves** — nonparametric descriptive baseline.
2. **Cox proportional hazards models** — semi-parametric, with and without CV indicators as time-dependent covariates.
3. **Weibull models** — fully parametric, enabling explicit calendar-time predictions.

CV indicators (mean IRI, within-week roughness variability, and mean vehicle count) were treated as time-dependent covariates, lagged one week to avoid simultaneity. Uncertainty was quantified via **section-level bootstrap resampling** ($B = 300$), preserving within-section dependence across recurrent deterioration episodes.

3. Key Results at a Glance

Finding	What It Means
CV data improves predictions	Adding connected-vehicle indicators significantly improved model fit ($p < 0.001$).
Weibull gives usable predictions	Median time-to-maintenance: 17.7 weeks (95% CI: 16.9–18.7 weeks).
Cox model falls short	Cox could not produce a usable median within the study period—model choice matters.
Uncertainty is small but real	Predictions are stable (2.7% relative uncertainty), giving planners confidence.

4. Key Findings in Detail

1. CV indicators improve predictive performance

Adding CV indicators to the Cox model significantly improved fit (LRT $\chi^2(2) = 34.48$, $p < 0.001$). The vehicle count indicator (`r_count_mean`) showed a protective association (HR = 0.73, $p < 0.001$)—sections with higher CV coverage reach the intervention threshold later, likely reflecting prioritization or monitoring intensity rather than intrinsic road quality.

Key Finding: CV-derived indicators add statistically significant explanatory value beyond static road attributes.

2. Model specification fundamentally shapes predictions

The Cox model could not estimate a median time-to-maintenance within the study period because the reference survival curve remained above 0.5 throughout follow-up. In contrast, the Weibull model (shape $\hat{\gamma} = 0.694$, indicating decreasing hazard) produced a **median time-to-maintenance of 17.71 weeks**.

Key Finding: Weibull yields finite, interpretable calendar-time medians where Cox cannot—directly affecting planning outputs.

3. Parameter uncertainty is modest but decision-relevant

Bootstrap analysis yielded a **95% CI of [16.87, 18.65] weeks** (SD = 0.482 weeks; relative uncertainty $\approx 2.7\%$). The bootstrap distribution was approximately normal and well-behaved.

Key Finding: Section-level bootstrap produces compact, interpretable intervals suitable for risk-aware scheduling.

5. Practical Implications

- **CV data adds value** beyond static road inventories for identifying elevated-risk sections.
- **Uncertainty-aware planning is feasible:** lower confidence bounds (e.g., ~ 16.9 weeks) support conservative scheduling, while interval widths identify where further monitoring would most reduce decision uncertainty.
- **Model choice should match operational goals:** Cox models suit relative-risk assessment and

covariate screening; Weibull (or other parametric) models provide the calendar-time predictions needed for budgeting and planning.

6. Skills Demonstrated

Skill Area	Specific Techniques
Survival Analysis	Cox PH, Weibull, Kaplan–Meier
Time-Dependent Covariates	Counting-process (start–stop) structure
Uncertainty Quantification	Section-level bootstrap, confidence intervals
Model Diagnostics	Schoenfeld residuals, AIC, likelihood-ratio tests
Statistical Computing	R, <code>survival</code> package, <code>data.table</code>
Domain Application	Infrastructure management, connected vehicle data

7. Limitations

- Findings are based on two Swedish municipalities; generalizability to other regions or climates is uncertain.
- Measurement error in CV indicators was not explicitly modeled, so prediction intervals likely understate total uncertainty.
- Within-section dependence across recurrent episodes was treated as conditionally independent, potentially understating correlation.
- Residual non-proportionality persists in the Cox models for some covariates.

8. Main Conclusion

Uncertainty quantification is essential for risk-aware pavement management. This thesis demonstrates that integrating high-frequency CV indicators into survival models combined with section-level bootstrap resampling enables a shift from deterministic scheduling to probabilistic, uncertainty-aware planning, supporting more informed and efficient maintenance decisions.

Full Thesis and Reproducible Code:

github.com/your-username/pavement-deterioration-survival-modeling

Keywords: Bootstrap · Cox proportional hazards · International Roughness Index · Parametric models · Pavement Management System · Semi-parametric models · Weibull distribution